Education Readiness Risk Modeling in R

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Recommendation

Use **Piecewise readiness** as an interpretable next-assessment support-risk model for public-safe education planning. The model should rank students for human review and support planning; it should not automate academic decisions.

The selected model achieved holdout AUC **0.938**, log loss **0.309**, and Brier score **0.093**. Repeated cross-validation produced log loss **0.318** and AUC **0.935**.

At a 50% support-review threshold, the workflow flags 326 holdout student transitions (48.9%), captures 298 of 347 observed support-risk cases, and produces PPV 91.4%. This is an operating example; capacity and intervention policy should set the final threshold.

The ranking view is valuable even without a single cutoff: the highest-risk decile has 1.92x lift over the base rate, and the top two deciles capture 38.3% of observed support-risk cases.

Direct Answers

1. The primary modeled outcome is next-window support risk, defined as a next assessment score below 50 or next-window nonparticipation. The holdout event rate is 52.1%.
2. The best operating model is **Piecewise readiness**, selected from interpretable GLM candidates after comparing nonlinear and benchmark model families.
3. The mathematical discovery step matters: Nonparametric smoothing supported a threshold-like readiness curve; piecewise and polynomial candidates were tested against spline and periodic benchmarks.
4. The model is strongest as a prioritization tool. Thresholds convert probabilities into workload, missed-risk, and precision tradeoffs.
5. The analysis is public-safe: it uses simulated identifiers and generalized assessment behavior, and excludes private prompts, exams, real student-identifiable records, credentials, and private source documents.

Data Audit

The analysis uses a public-safe assessment extract with one row per assessment window. The modeling table turns consecutive assessment windows into prediction records: current assessment information is used to predict support risk at the next assessment.

Measure	Value
Raw assessment rows	4,018
Modeled transitions	3,322
Unique public-safe student IDs	696
Support-risk event rate	52.1%
Current nonparticipation rate	7.2%
Next-window nonparticipation rate	7.2%
Median current readiness	47.7
Included assessment windows	beginning-of-year, end-of-year

The data is public-safe by design. Student, teacher, section, and course identifiers are simulated labels; score/readiness behavior is generalized from a bootstrapped assessment workflow and should not be treated as a real student-record extract.

Model Journey

The model search followed the same statistical logic used in the private MS statistics work, but with an original public-safe education framing: inspect the shape first, test candidate parametric families second, and keep the final model interpretable unless a flexible benchmark clearly earns its complexity.

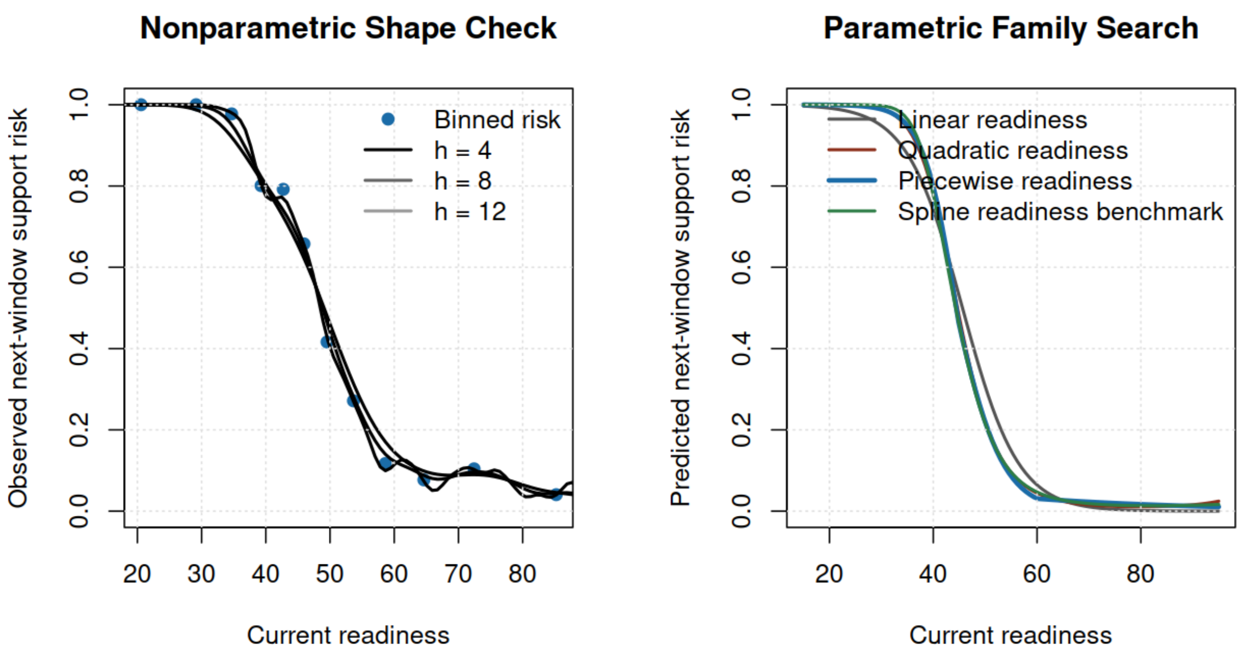

Figure 1: Nonparametric and parametric shape discovery

Family	Why tested	Decision	CV loss	Holdout AUC
Context baseline	Tests whether demographic and operating context alone is enough.	Rejected; validation is much weaker without readiness.	0.643	0.673
Linear readiness	Adds the main readiness signal with a simple monotone probability shape.	Rejected; ranking is strong but probability quality is worse.	0.340	0.933
Quadratic readiness	Tests whether risk accelerates near low readiness values.	Rejected; close to selected model, but less directly aligned with the discovered threshold shape.	0.319	0.937
Cubic polynomial readiness	Checks whether a more flexible polynomial improves fit enough to justify instability risk.	Rejected; added polynomial curvature without improving the operating story.	0.318	0.936
Piecewise readiness	Uses the smooth shape discovery to encode separate readiness regions.	Selected as the operating model.	0.318	0.938

Family	Why tested	Decision	CV loss	Holdout AUC
Periodic context benchmark	Tests recurring assessment-window structure without making periodicity the headline.	Benchmark only; periodic terms did not justify replacing the operating model.	0.320	0.937
Spline readiness benchmark	Flexible nonlinear benchmark for the readiness curve.	Benchmark only; used to test whether flexible curvature changes the conclusion.	0.318	0.937

Candidate logistic models were then compared with repeated stratified 5-fold cross-validation on the training split. Log loss is the primary criterion because a support-prioritization workflow needs useful probabilities, not only rank ordering.

Model	Selected	Role	Params	CV loss	CV SD	CV AUC	Holdout loss	Holdout AUC	Delta
Piecewise readiness	Yes	Selection candidate	15	0.318	0.001	0.935	0.309	0.938	0.000
Cubic polynomial readiness		Selection candidate	15	0.318	0.001	0.935	0.313	0.936	0.000
Spline readiness benchmark		Benchmark	16	0.318	0.001	0.935	0.313	0.937	0.000
Quadratic readiness		Selection candidate	14	0.319	0.001	0.935	0.316	0.937	0.001
Periodic context benchmark		Benchmark	16	0.320	0.001	0.935	0.315	0.937	0.002
Linear readiness		Selection candidate	13	0.340	0.001	0.935	0.358	0.933	0.022
Context baseline		Selection candidate	10	0.643	0.000	0.674	0.644	0.673	0.325

The selection rule favors the simplest non-benchmark model within one standard error of the best repeated-CV log loss. That rule protects the portfolio story from choosing a visually impressive model that does not materially improve validated probability quality.

Final Model

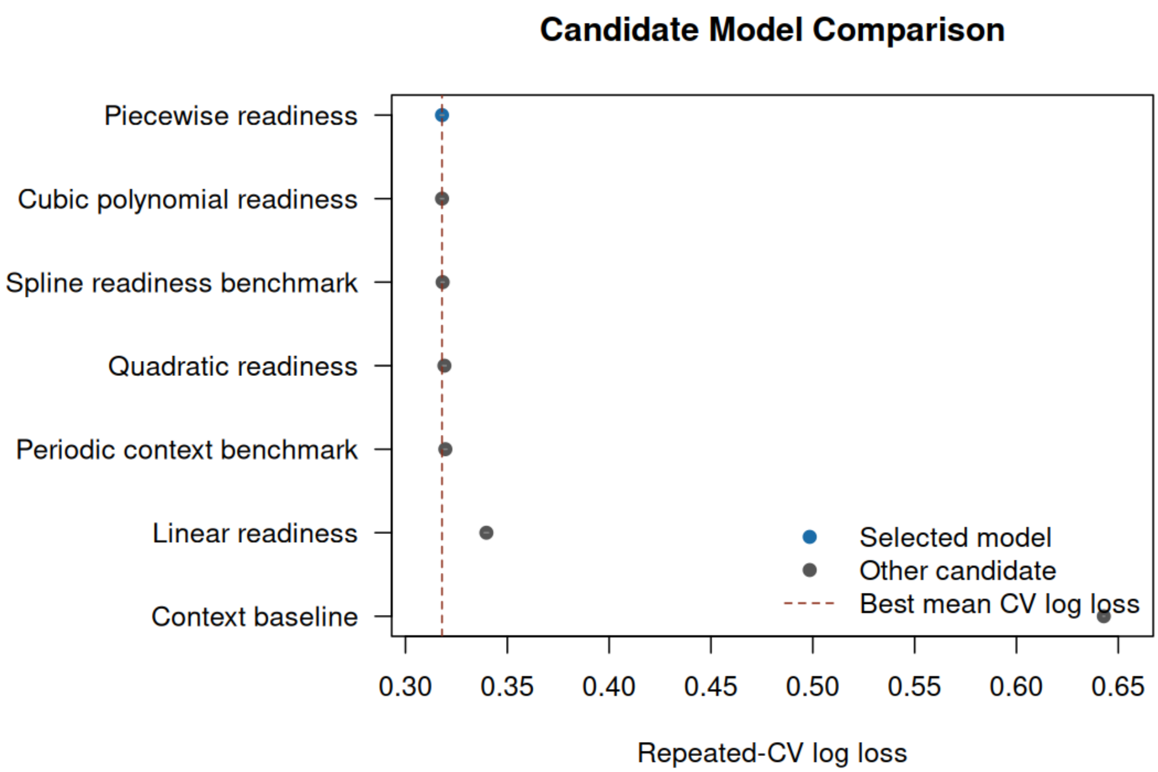

Figure 2: Candidate model comparison

Metric	Value
Selected model	Piecewise readiness
Selection rule	Simplest non-benchmark model within one standard error of best repeated-CV log loss
Shape discovery result	Nonparametric smoothing supported a threshold-like readiness curve; piecewise and polynomial candidates were tested against spline and periodic benchmarks.
Training rows	2656
Holdout rows	666
Training event rate	52.1%
Holdout event rate	52.1%
Repeated CV folds	5
Repeated CV repeats	6
CV log loss	0.318
CV AUC	0.935
Holdout log loss	0.309
Holdout Brier score	0.093
Holdout AUC	0.938
Calibration intercept	0.165
Calibration slope	1.025
Training log loss	0.312
Training AUC	0.938

Bootstrap intervals give a practical uncertainty band around the holdout metrics.

Metric	Estimate	95% CI
LogLoss	0.309	0.268 to 0.352
Brier	0.093	0.079 to 0.107
AUC	0.938	0.919 to 0.955

The adjusted odds ratios below translate the selected GLM into stakeholder-readable effects.

Predictor	Scale	Odds ratio	95% CI	p-value
Grade 10 vs grade 9	1-unit / level change	1.03	0.74 to 1.44	0.869
Grade 11 vs grade 9	1-unit / level change	1.11	0.78 to 1.57	0.576
Grade 12 vs grade 9	1-unit / level change	1.08	0.65 to 1.80	0.770
Honors track vs regular	1-unit / level change	1.27	0.85 to 1.91	0.242
AP track vs regular	1-unit / level change	1.25	0.91 to 1.73	0.173
Beyond-core track vs regular	1-unit / level change	2.78	0.69 to 11.29	0.152
Current window: end of year	1-unit / level change	5.62	4.18 to 7.56	<0.001
Attendance category: high absence vs normal	1-unit / level change	0.83	0.57 to 1.23	0.363
Attendance category: at-risk absence vs normal	1-unit / level change	1.26	0.41 to 3.88	0.689
Attendance probability	0.1-unit change	0.80	0.50 to 1.29	0.366
Readiness shortfall below 45	5-unit change	4.73	3.60 to 6.23	<0.001

Predictor	Scale	Odds ratio	95% CI	p-value
Readiness gain from 45 to 60	5-unit change	0.34	0.29 to 0.39	<0.001
Readiness gain above 60	5-unit change	0.86	0.75 to 0.98	0.023
School-year sequence	1-unit / level change	0.97	0.91 to 1.04	0.378

Probability Scale

Risk categories provide a bridge between calibrated probabilities and support workflows.

Category	Students	Share	Pred	Obs	Cases
Monitor	298	44.7%	11.0%	11.1%	33
Watch	42	6.3%	41.1%	38.1%	16
Review	43	6.5%	56.4%	72.1%	31
Priority	283	42.5%	92.5%	94.3%	267

A threshold turns probabilities into a work queue. Lower thresholds catch more support-risk cases but create more reviews; higher thresholds focus capacity but miss more students.

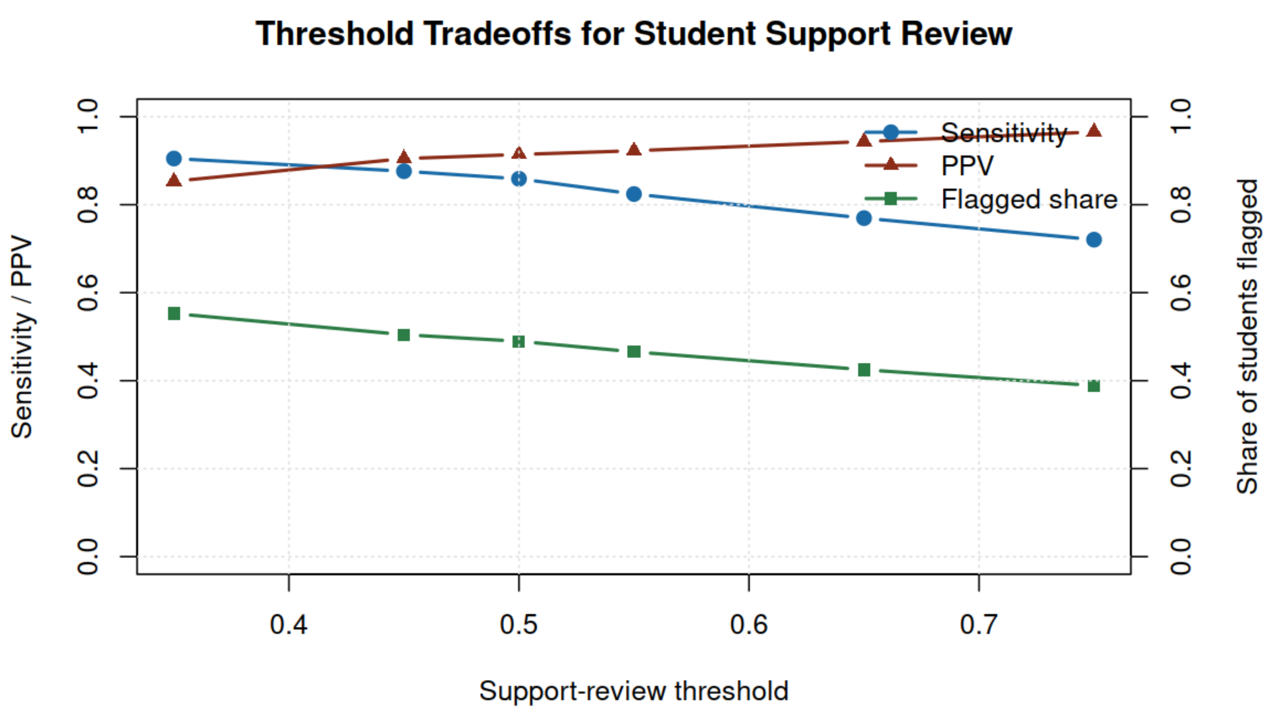

Figure 3: Threshold tradeoffs

Threshold	Flagged	Flagged %	Captured	Sens	Spec	PPV	NPV
35%	368	55.3%	314 of 347	90.5%	83.1%	85.3%	88.9%
45%	336	50.5%	304 of 347	87.6%	90.0%	90.5%	87.0%
50%	326	48.9%	298 of 347	85.9%	91.2%	91.4%	85.6%

Threshold	Flagged	Flagged %	Captured	Sens	Spec	PPV	NPV
55%	310	46.5%	286 of 347	82.4%	92.5%	92.3%	82.9%
65%	283	42.5%	267 of 347	76.9%	95.0%	94.3%	79.1%
75%	259	38.9%	250 of 347	72.0%	97.2%	96.5%	76.2%

The table below uses illustrative support-planning economics to show how a threshold can be chosen from capacity and intervention assumptions. These values are scenario assumptions, not claims about a real school system.

Threshold	Flagged	Captured	Benefit	Cost	Net
35%	368	314	$70,650	$27,600	$43,050
45%	336	304	$68,400	$25,200	$43,200
50%	326	298	$67,050	$24,450	$42,600
55%	310	286	$64,350	$23,250	$41,100
65%	283	267	$60,075	$21,225	$38,850
75%	259	250	$56,250	$19,425	$36,825

Model Checks

ROC checks ranking quality. Calibration checks whether predicted probabilities are on the right scale across ordered risk bands.

Band	N	Pred	Obs	Expected	Cases
Band 1	83	2.2%	8.4%	1.8	7
Band 2	83	6.3%	9.6%	5.2	8
Band 3	83	14.7%	9.6%	12.2	8
Band 4	84	32.8%	25.0%	27.6	21
Band 5	83	62.0%	73.5%	51.4	61
Band 6	83	87.5%	90.4%	72.6	75
Band 7	83	98.2%	100.0%	81.5	83
Band 8	84	99.9%	100.0%	83.9	84

Diagnostic	Estimate	Interpretation
Calibration intercept	0.165	Near 0 means predicted risk is not systematically high or low
Calibration slope	1.025	Near 1 means predicted probabilities are not overly extreme or compressed

Subgroup calibration checks show where monitoring would matter before operational use. The table reports groups with at least 25 holdout records.

Group	Level	N	Pred	Obs	Gap	Cases
course track	ap	211	33.7%	39.3%	5.7%	83
course track	honors	84	33.9%	29.8%	-4.1%	25
course track	regular	368	64.0%	64.7%	0.6%	238
assessment window	beginning-of-year	405	45.4%	49.9%	4.5%	202
assessment window	end-of-year	261	58.4%	55.6%	-2.9%	145

Group	Level	N	Pred	Obs	Gap	Cases
attendance category	at-risk	60	53.7%	63.3%	9.6%	38
attendance category	high	285	49.6%	49.5%	-0.1%	141
attendance category	normal	321	50.7%	52.3%	1.7%	168

A ranked queue is often more useful than a single classification cutoff. The lift chart shows how concentrated support-risk cases are in the highest predicted-risk deciles.

Decile	N	Pred	Obs	Cases	Lift	Capture
1	66	99.9%	100.0%	66	1.92x	19.0%
2	67	99.1%	100.0%	67	1.92x	38.3%
3	66	95.2%	98.5%	65	1.89x	57.1%
4	67	82.0%	86.6%	58	1.66x	73.8%
5	67	58.8%	70.1%	47	1.35x	87.3%
6	66	35.2%	24.2%	16	0.47x	91.9%
7	67	18.8%	17.9%	12	0.34x	95.4%
8	66	10.1%	6.1%	4	0.12x	96.5%
9	67	4.4%	11.9%	8	0.23x	98.8%
10	67	2.0%	6.0%	4	0.11x	100.0%

Sensitivity Check

The sensitivity analysis lowers the support-risk score cut point from 50 to 45 and refits the selected model family. This tests whether the prioritization story depends on one particular threshold definition.

Measure	Primary	Sensitivity
Primary holdout event rate	52.1%	Reference
Sensitivity holdout event rate	Reference	41.0%
Sensitivity holdout log loss	0.309	0.329
Sensitivity holdout Brier score	0.093	0.098
Sensitivity holdout AUC	0.938	0.923
Rank correlation with primary predictions	Reference	0.989
Top-quintile overlap with primary ranking	Reference	97.8%
Students changing risk category	Reference	159 of 666

Scenario Profiles

Scenario profiles translate the model into concrete, public-safe support-planning examples with probability intervals.

Scenario	Grade	Track	Window	Attendance	Readiness	Risk	95% CI	Category
On-track check-point	10	regular	beginning-of-year	normal	72.0	2.1%	1.3% to 3.6%	Monitor
Attendance watch	9	regular	beginning-of-year	high	51.0	19.3%	10.1% to 33.7%	Monitor
Priority support	11	honors	end-of-year	at-risk	39.0	99.0%	97.9% to 99.5%	Priority

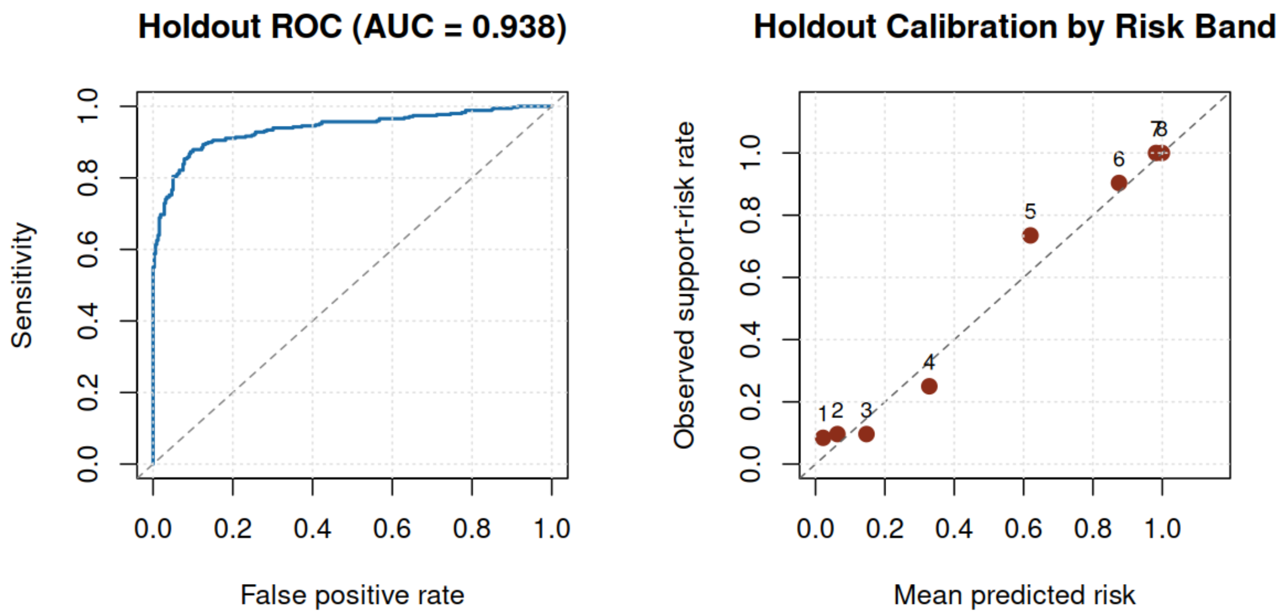

Figure 4: ROC and calibration diagnostics

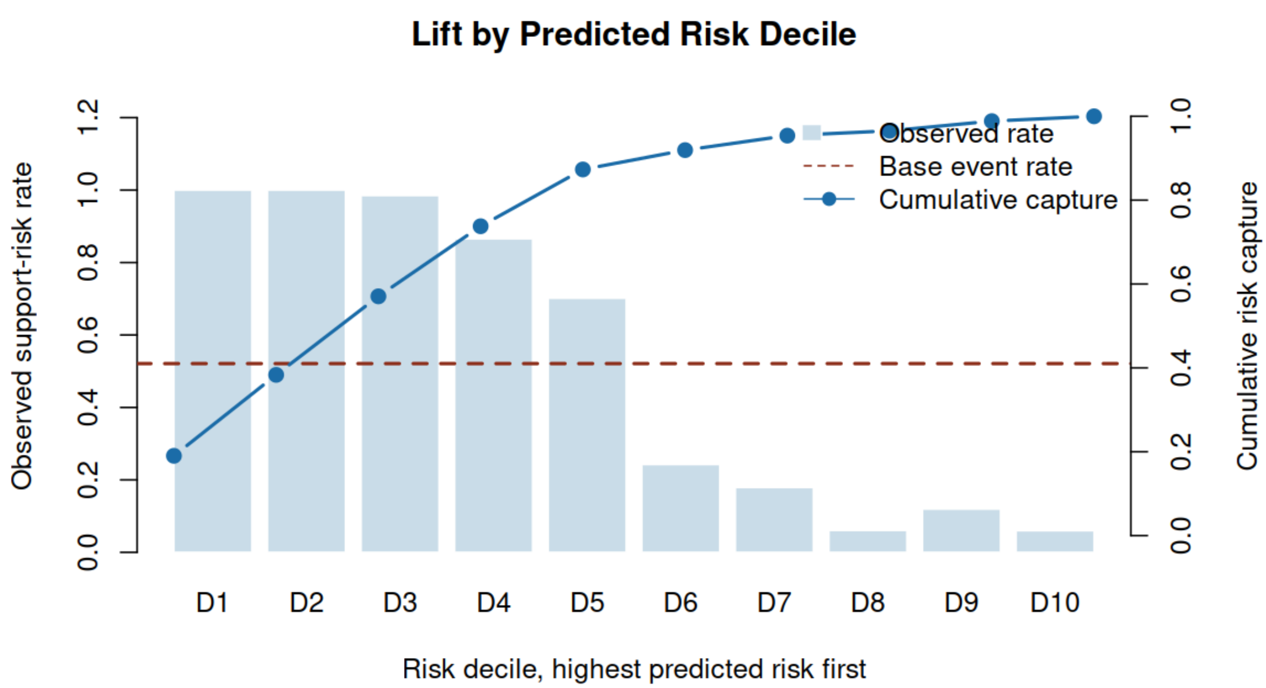

Figure 5: Lift by predicted risk decile

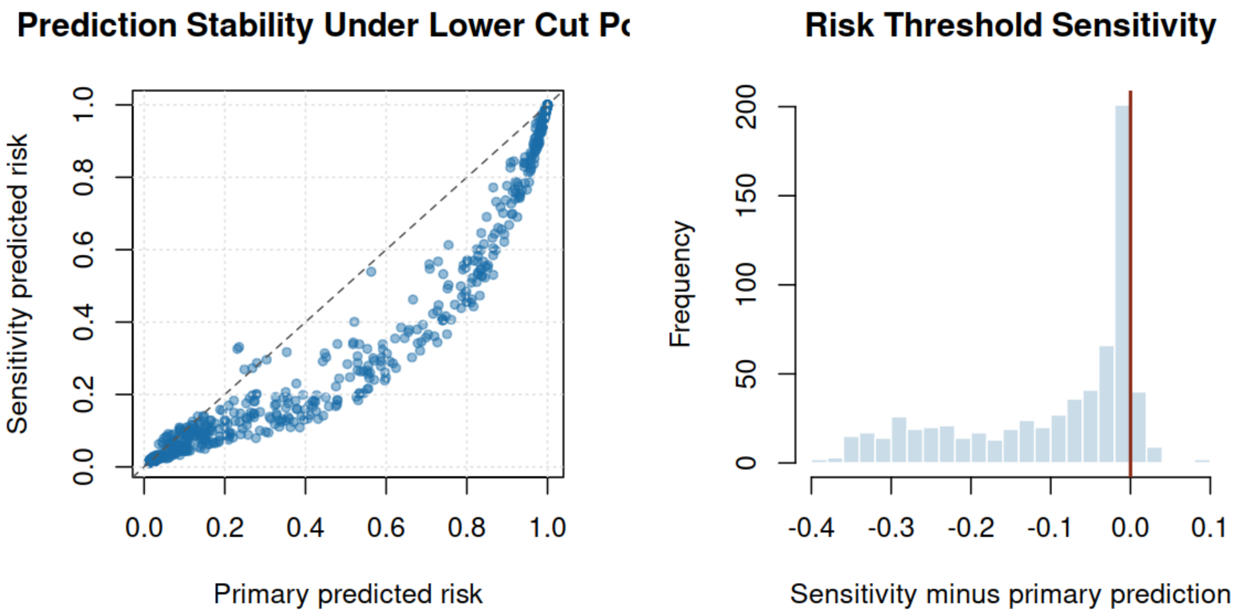

Figure 6: Sensitivity analysis

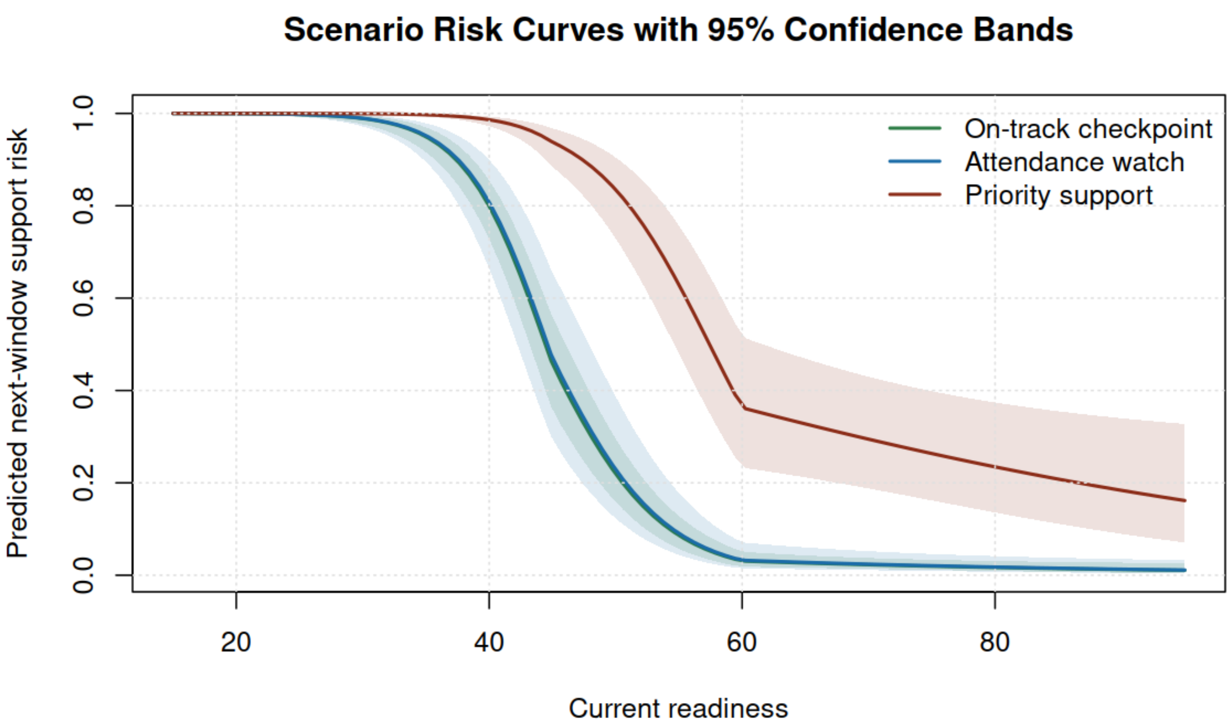

Figure 7: Scenario readiness risk curves

Bottom Line

- Use the model to prioritize human review and early support planning, not to automate student-level decisions.
- Keep the piecewise readiness shape because it captures the discovered nonlinear risk pattern while staying easier to explain than a flexible spline.
- Choose operating thresholds from review capacity, support cost, and tolerance for missed support-risk cases.
- Monitor calibration by course track, assessment window, and attendance group before treating risk categories as stable operating labels.

Reproducibility

Rebuild the full evidence packet with `make all`. The core pipeline uses base R, the included public-safe extract, and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, patient data, school-private records, credentials, and copyrighted source documents.